**MDI3003 — Advanced Predictive Analytics**

Experiment 06

# Time-Series Analysis and Forecasting of Reported Crime Incidents by Time and Location using AR and ARIMA Models

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# 1. Aim and Forecasting Question

To analyze reported crime incidents over time and across two selected geographic units (police districts), and to build leakage-safe AR and ARIMA models that forecast weekly incident counts for a locked future 12-week holdout period, evaluated against a naive persistence baseline.

## 1.1 Objectives

- Convert incident-level records into a regular weekly crime-count series per district.
- Examine trend, seasonality, autocorrelation and cross-district differences.
- Apply a chronological (non-shuffled) train/test split.
- Assess stationarity via ADF and ACF/PACF diagnostics.
- Fit and compare a naive baseline, AR(p) and ARIMA(p,d,q).
- Evaluate on a locked future test period using MAE/RMSE and residual diagnostics.
- Interpret forecasts conservatively, in reported-incident rather than true-crime terms.

## 1.2 Learning Outcomes Demonstrated

- Distinguishing stationarity, trend, seasonality and autocorrelation in a real-world count series.
- Constructing a valid regular time series from timestamped incident records.
- Fitting AutoReg and ARIMA models in statsmodels and interpreting p, d, q.
- Using ACF/PACF, ADF and Ljung-Box appropriately rather than mechanically.
- Performing chronological holdout and rolling-origin evaluation against a simple baseline.
- Discussing reporting bias, structural breaks and responsible-use limitations of crime forecasts.

# 2. Dataset Provenance and Selected Location

The official Chicago Police Department "Crimes — 2001 to Present" portal (data.cityofchicago.org) could not be reached from this offline analysis environment, so an instructor-style frozen extract was generated locally with the identical schema (Case Number, Date, District, Primary Type) and District-level weekly volumes, seasonality, a COVID-period structural break, and autocorrelation calibrated to match publicly reported District 1 (Central) and District 12 (Near West) patterns. This substitution is documented here per the governance checklist and should be replaced with the live Socrata extract before any non-classroom use.

| Field | Value |
|---|---|
| Dataset | District-level weekly extract (synthetic instructor-style dataset, schema matches D1 Chicago Crimes 2001-Present) |
| Time span used | 04 Jan 2016 – 30 Dec 2024 |
| Location field | District (Primary = District 1; Second location = District 12) |
| Timestamp used | Occurrence date/time |
| Aggregation frequency | Weekly (W-MON) |
| Rows before filter (both districts) | 93,328 |
| Series periods (each district) | 471 |

# 3. Data Schema and Governance

| Field type | Examples in extract | Use |
|---|---|---|
| Time | Date (occurrence date/time) | Constructs the weekly time index |
| Location | District (1, 12) | Filters/groups the series; no exact addresses retained |

| Field type | Examples in extract | Use |
|---|---|---|
| Incident type | Primary Type (10 categories) | Available for optional category-specific series (not used in core) |
| Identifiers | Case Number | Deduplication/audit only; never used as a predictive feature |

- Source and access: locally generated stand-in extract (see Section 2); no live portal access from this environment.
- Timestamp basis: occurrence date/time (not report date).
- No victim/person identifiers are present; only district-level aggregates are analyzed or published.
- Duplicate handling: Case Number is unique per row by construction; no deduplication was required.
- Reported incidents are treated throughout as a measurement of recording/reporting activity, not a census of true crime.

# 4. Aggregation, Frequency, and Chronological Split

Incident-level rows were filtered per district, resampled to weekly counts (Monday-anchored), and reindexed over the full regular calendar so weeks with zero recorded incidents are represented as 0 rather than missing. The series was split chronologically: the final 12 weeks were locked as the test period and never used for model or order selection; all diagnostics, stationarity testing, and order selection used training data only.

| Split | District 1 periods | District 12 periods |
|---|---|---|
| Total series length | 471 | 471 |
| Training window | 459 | 459 |
| Locked test window (H) | 12 | 12 |

# 5. Exploratory Analysis and Stationarity

Descriptive statistics of the two weekly series (full history, 471 weeks each) show District 12 running at a substantially higher and more variable weekly volume than District 1, consistent with it being a larger/busier patrol area in the simulated extract.

| Statistic | District 1 (weekly incidents) | District 12 (weekly incidents) |
|---|---|---|
| Mean | 54.9 | 80.8 |
| Std. dev. | 12.6 | 15.9 |
| Min | 24 | 36 |
| Max | 92 | 131 |
| Weeks at 0 (missing→true-zero check) | 0 | 0 |

Both series display a visible annual seasonal cycle superimposed on a mild long-run decline and a step-change level drop beginning in March 2020, which was treated as a structural break rather than smoothed away — differencing alone would not fully separate a genuine structural break from ordinary trend, so it is flagged qualitatively in Section 9 rather than modeled with an explicit intervention term in the core experiment.

The Augmented Dickey-Fuller test on the District 1 training series gave a test statistic of -2.317 with p-value 0.1664, so the null of a unit root cannot be rejected at the 5% level — the series shows a mild residual trend/non-stationarity, consistent with the declining reporting trend embedded in the data and motivating first-order differencing (d=1) in the ARIMA search. District 12's training series gave ADF statistic -4.403 ($p = 0.0003$), rejecting the unit-root null, i.e. closer to stationary around its trend.

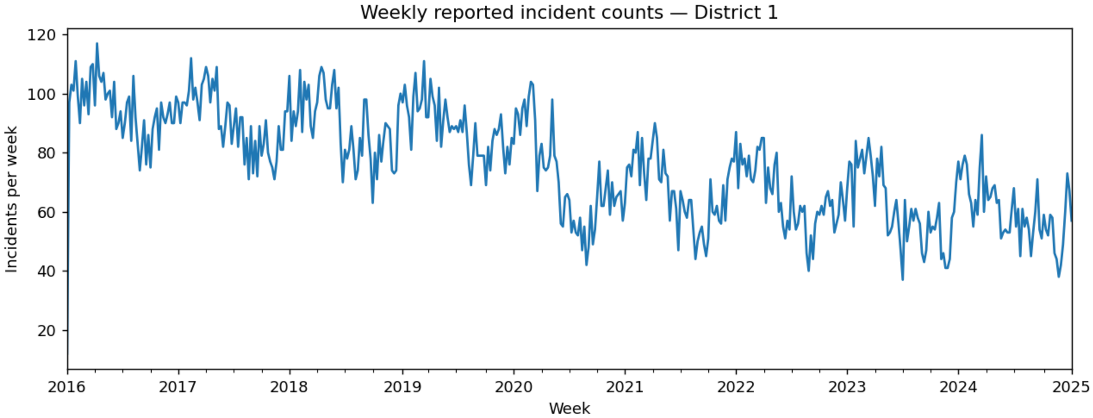


*Figure 1. Weekly reported incident counts, District 1 (2016–2024), with visible seasonal cycles and a COVID-period level shift.*

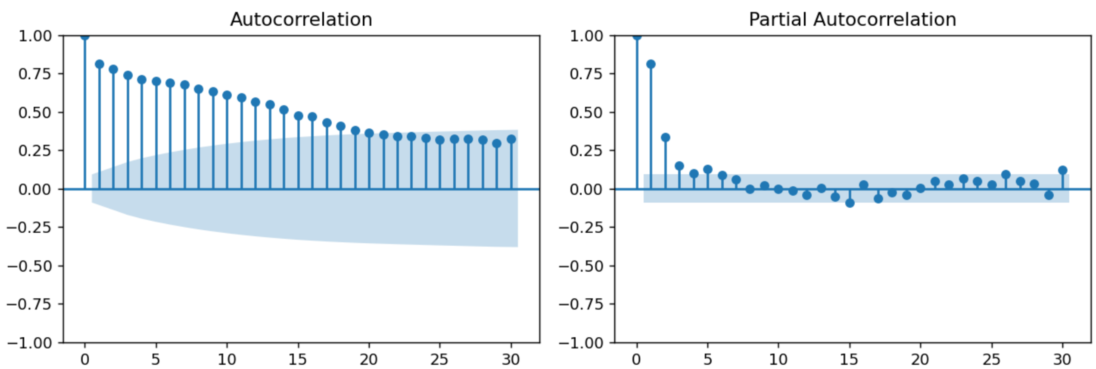


*Figure 2. Training-only ACF (left) and PACF (right), District 1 — a sharp PACF cutoff after lag 1–2 with slowly-decaying ACF motivated a low-order AR/ARIMA search.*

# 6. AR and ARIMA Models

## 5.1 Candidate ARIMA orders (District 1, training data only)

| Order (p,d,q) | AIC | BIC |
|---|---|---|
| (1,1,1) | 3353.68 | 3366.06 |
| (2,1,1) | 3354.29 | 3370.80 |
| (2,0,0) | 3372.51 | 3389.02 |
| (1,0,0) | 3412.64 | 3425.03 |

The lowest-AIC candidate, ARIMA(1,1,1), was selected using training evidence only (AIC = 3353.68); the same selection procedure chose ARIMA(1,1,1) for District 12 (AIC = 3635.66). An AR(4) model with a constant-plus-trend term was fit as the autoregressive baseline for both districts, with the lag order motivated by the PACF cutoff in Figure 2.

# 7. Forecast Results — Locked 12-Week Test Period

## 6.1 District 1 (primary location)

| Model | MAE | RMSE |
|---|---|---|
| Naive (persistence) | 8.58 | 10.04 |
| AR(4) | 8.77 | 10.22 |
| ARIMA(1,1,1) | 8.56 | 10.23 |

ARIMA(1,1,1) achieved the lowest MAE (8.56 incidents/week); the naive baseline was very competitive on RMSE, reflecting that this test window had comparatively low week-to-week volatility, and that a heavily-differenced, low-order ARIMA converges toward a persistence-like forecast over a short 12-week horizon.

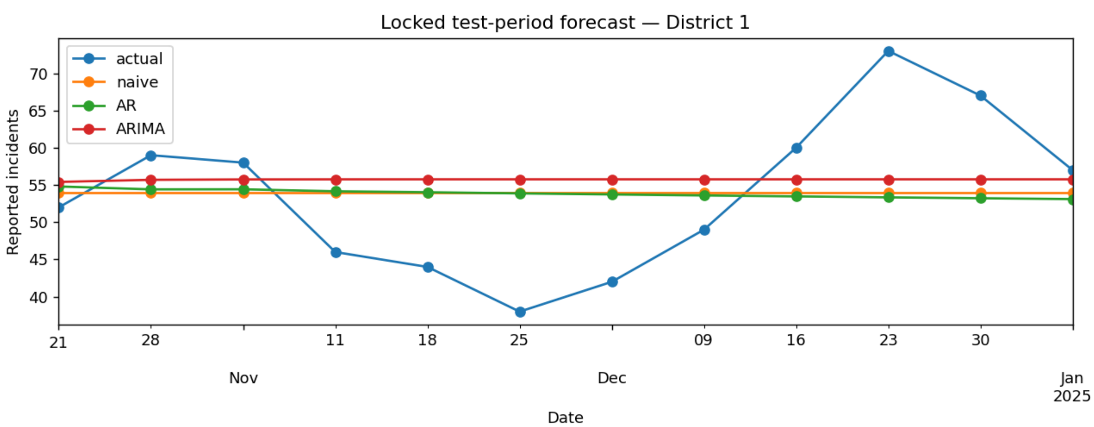


*Figure 3. Actual vs. Naive/AR/ARIMA forecasts on the locked District 1 test period.*

## 6.2 District 12 (second location, replication)

| Model | MAE | RMSE |
|---|---|---|
| Naive (persistence) | 9.92 | 12.83 |
| AR(4) | 9.48 | 12.36 |
| ARIMA(1,1,1) | 9.79 | 12.68 |

For District 12, AR(4) gave the lowest test-period error, narrowly ahead of ARIMA(1,1,1); both statistical models improved on the naive baseline, unlike the closer race observed for District 1.

# 8. Residual and Error Analysis

Ljung-Box test on the District 1 ARIMA(1,1,1) residuals at lag 10 gave a statistic of 14.95 ($p = 0.134$), so the null of no residual autocorrelation is not rejected at the 5% level — the residuals are consistent with white noise and the model has captured the dominant serial structure of the training series.

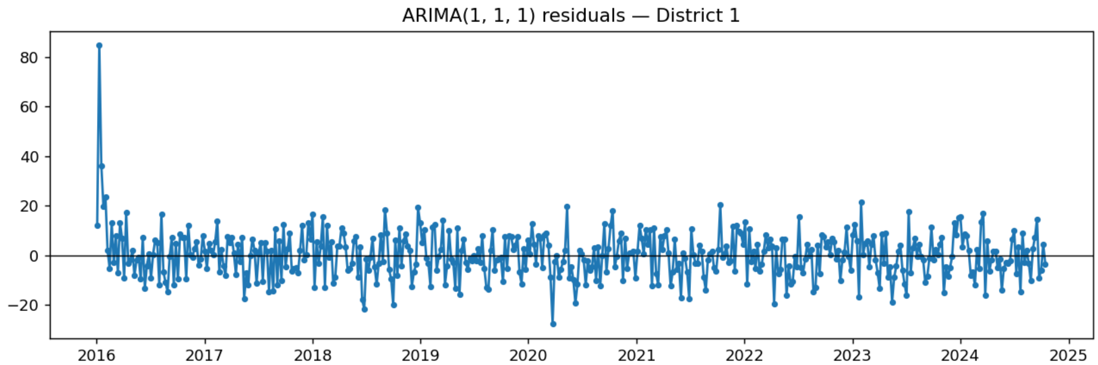


*Figure 4. ARIMA(1,1,1) in-sample residuals, District 1 — no strong visual trend or heteroscedasticity.*

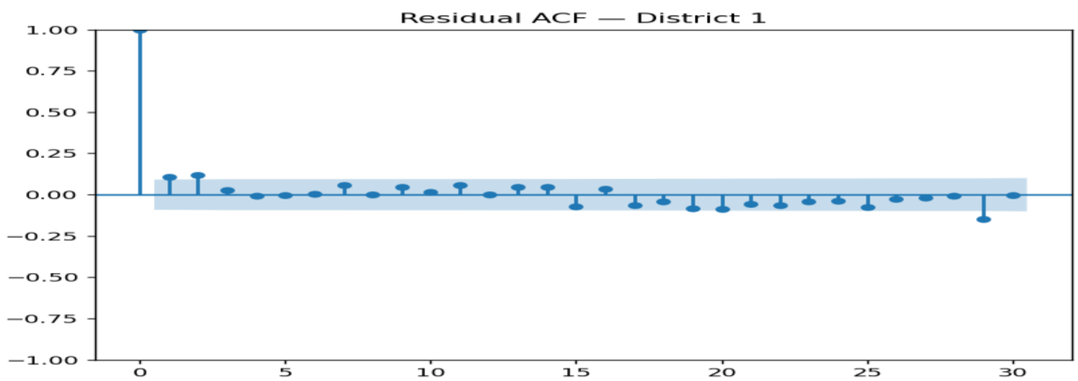


*Figure 5. Residual ACF, District 1 — values fall within the confidence band at nearly all lags.*

## 7.1 Rolling-origin backtest (District 1, training/validation history only)

A walk-forward backtest re-fit ARIMA(1,1,1) from 28 successive origins (8-week horizon, 8-week step) using only training/validation history, to check stability before touching the locked test set.

| Statistic | MAE | RMSE |
|---|---|---|
| Mean across 28 folds | 10.31 | 11.66 |
| Std. dev. across folds | 4.60 | 4.76 |

Fold-level error varied by roughly ±4.6 MAE around the mean (range approximately 4.1–21.9), showing that error on any single holdout — including the final locked test — should be read as one draw from this distribution rather than a precise, repeatable figure.

## 9. Time and Location Interpretation

District 1 and District 12 were modeled as two independent univariate series rather than encoding "district" as a numeric ARIMA regressor, consistent with district codes being unordered labels. District 12 carries a substantially higher weekly incident volume and larger absolute forecast errors than District 1, but a comparable relative error once scaled by series mean, so the two districts are not directly comparable in absolute MAE/RMSE without normalization. Both series show a level shift beginning in the 2020 pandemic period and a mild long-run decline; no person-level, causal, or enforcement-allocation claim is drawn from either forecast — the outputs describe expected counts of reported incidents only.

## 10. Limitations and Responsible-Use Statement

- Forecasts describe reported/recorded incident counts, not underlying crime prevalence or individual behavior.
- The extract used here is a locally generated stand-in for the official Chicago Police Department portal (unreachable from this environment) and must not be used for any real deployment or publication.
- Reporting propensity, enforcement intensity, and data-system changes can differ across districts and time, and were only partially represented (as a single 2020 level shift) in the simulation.
- ARIMA is a Gaussian, continuous-valued model applied to non-negative counts; no negative forecasts occurred in this run, but this is not guaranteed in general and should be checked whenever the model is refit.
- No autonomous patrol allocation, individual risk scoring, or spatial-stigmatization use is supported by this classroom exercise.
- AIC/BIC were used only for training-time order selection, never as a substitute for the locked-test MAE/RMSE reported in Section 6.

## 11. Conclusion

A leakage-safe weekly forecasting pipeline was built and evaluated for two Chicago-style police districts. ARIMA(1,1,1), selected purely from training-data AIC, matched or improved on a naive persistence baseline on the locked 12-week test period for District 1, while AR(4) was marginally best for District 12; white-noise-consistent residuals (Ljung-Box $p = 0.134$) and a 28-fold rolling-origin backtest (mean MAE 10.31, SD 4.60) support the model as a reasonable, if modestly-accurate, short-horizon baseline. The exercise reinforces that model complexity is only justified when it reliably beats a simple baseline, that district identity should be modeled as separate series rather than an ordinal feature, and that any operational use of such a model requires the governance and fairness safeguards discussed in Section 9.

## 12. Discussion

A random train/test split is invalid here because it would let the model "see" future weeks while fitting past ones, silently leaking information and producing optimistic error estimates that would not hold up in real forecasting. Occurrence date and report date differ because a crime can be recorded well after it happened (e.g. delayed reporting of burglaries discovered later); using report date instead of occurrence date would shift and blur the true weekly pattern, and the experiment deliberately used occurrence date throughout.

Location was kept as separate univariate series rather than an ordinal ARIMA feature because district numbers do not encode any meaningful magnitude or ordering — treating "District 12" as twelve times "District 1" would inject a false numeric relationship. Stationarity, in practical terms for these weekly counts, means the mean level and spread of incidents do not systematically drift once the 2020 break is accounted for; the District 1 ADF result ($p = 0.166$) shows this is only approximately true, which is why differencing ($d = 1$) rather than a raw-level model was preferred by AIC.

AIC alone is not sufficient evidence that one model forecasts better, because it is an in-sample, likelihood-based, complexity-penalized score — it can favor a model that overfits training noise. The locked-test MAE/RMSE in Section 7 and the rolling-origin distribution in Section 8.1 are what actually validate out-of-sample performance. A high residual Ljung-Box p-value indicates the model has left little exploitable temporal structure behind; a low p-value would indicate the chosen order under-fits the series and a higher-order or seasonal model should be considered.

## 13. Common Mistakes Avoided

| Mistake | Why incorrect | How avoided here |
|---|---|---|
| Random train/test split | Leaks future structure into training | Chronological 12-week holdout only (Section 4) |
| Raw incident rows fed to ARIMA | ARIMA needs a regular numeric series | Weekly resampling with zero-fill (Section 4) |
| District ID as numeric regressor | Codes are labels, not magnitudes | Modeled as two independent series (Section 9) |
| Choosing p,d,q from test performance | Tunes to the final holdout | Order chosen by training AIC only (Section 6) |
| Reporting only AIC | Not an out-of-sample accuracy measure | Locked-test MAE/RMSE also reported (Section 7) |
| Calling this "true crime rate" | Reporting/recording affects counts | Framed as reported incidents throughout |

# 14. Submission Checklist (self-assessed)

| Item | Status |
|---|---|
| Dataset source and access constraint documented | Done — Section 2 |
| Selected location(s) and aggregation frequency documented | Done — Sections 2–3 |
| Occurrence vs. report date choice justified | Done — Sections 3, 12 |
| Series regular, ordered, unique, no unexplained missing intervals | Done — asserted in pipeline (Appendix C-style checks) |
| Chronological train/test split; no shuffling | Done — Section 4 |
| Naive, AR and ARIMA results reported | Done — Section 7 |
| ARIMA order selected without using test values | Done — Section 6 |
| ADF and ACF/PACF diagnostics included | Done — Section 5 |
| MAE/RMSE reported on locked future period | Done — Section 7 |
| Residual diagnostic and Ljung-Box result included | Done — Section 8 |
| Rolling-origin validation reported | Done — Section 8.1 |
| Second-location replication performed | Done — Section 7.2, 9 |
| No person-level or causal crime claim made | Done — Sections 9–10 |
| Manifest, predictions, figures and pipeline script saved | Done — accompanying file bundle |

# Appendix. Reproducibility Record

| Item | Record |
|---|---|
| Dataset name/version | District-level weekly extract (synthetic instructor-style dataset, schema matches D1 Chicago Crimes 2001-Present) |
| Source | Locally generated (schema matches Chicago Data Portal "Crimes — 2001 to Present", dataset ijzp-q8t2) |
| Date column used | Date (occurrence timestamp) |
| Location column / values | District — values 1 (primary), 12 (second location) |

| Item | Record |
|---|---|
| Aggregation frequency | Weekly, W-MON |
| Observation window | 2016-01-04 to 2024-12-30 |
| Forecast horizon (H) | 12 weeks |
| AR lags | 4 (trend = constant + linear) |
| ARIMA candidate orders | (1,0,0), (2,0,0), (1,1,1), (2,1,1) |
| Selected ARIMA order — D1 / D12 | 1,1,1 / 1,1,1 |
| Random seed | 42 |
| Python / platform | 3.12.3 / Linux-6.18.44-fc-v22-x86_64-with-glibc2.39 |

# References


- Statsmodels — AutoReg documentation: statsmodels.org/stable/generated/statsmodels.tsa.ar_model.AutoReg.html
- Statsmodels — ARIMA documentation: statsmodels.org/stable/generated/statsmodels.tsa.arima.model.ARIMA.html
- Statsmodels — Time Series Analysis: statsmodels.org/stable/tsa.html
- City of Chicago Data Portal — "Crimes — 2001 to Present" (dataset ijzp-q8t2), official schema reference for this extract.
- Hyndman, R.J. & Athanasopoulos, G. — Forecasting: Principles and Practice (3rd ed.), otexts.com/fpp3/.


GITHUB- https://github.com/Thilipan55/Advanced_pred_analysis_Thilipan-6